COM-402 Data Visualization: Milestone 2

Cinema Evolution Project

Project skeleton is available online at `https://com-480-data-visualization.github.io/DataFlix/`. The website currently displays only bar charts and scatter plots as proof-of-concept implementations to test which methods allowed us to connect the website with the data. Final visualizations will follow the specifications described in the report.

1 Introduction & Goal

Cinema is one of the most influential cultural industries, reflecting technological progress, societal changes, and evolving audience preferences. Over the past century, the film industry has undergone major transformations: from the rise of early black-and-white films to the dominance of global blockbusters and streaming platforms. These changes are not only artistic, but also structural, affecting production volume, genre diversity, audience reception, and financial scale.

Despite the richness of available movie datasets, understanding these long-term trends is challenging when data is presented in isolation. Static charts often fail to convey how different aspects of cinema interact and evolve together over time.

The goal of our project is to design an interactive visualization website that presents the evolution of cinema as a continuous, narrative-driven experience. Instead of showing disconnected visualizations, we structure the website as a chronological journey, where the user scrolls through time and progressively uncovers patterns and relationships in the data.

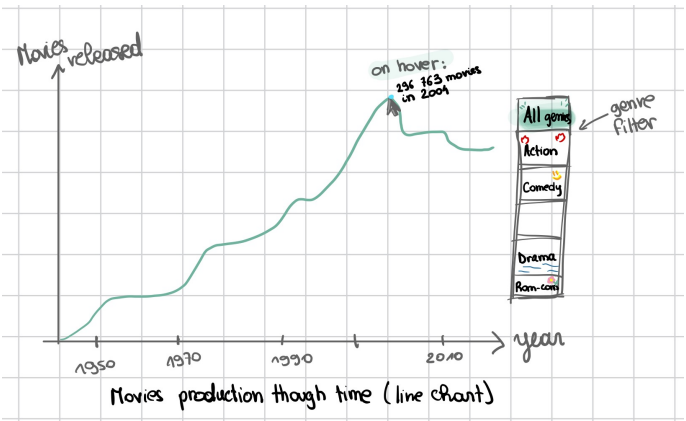
2 Sketches

2.1 Movie Production Over Time

Features: This visualization shows the number of movies released per year using a line chart (year on the x-axis, number of movies on the y-axis). A genre selection panel allows users to filter one or multiple genres, updating the chart dynamically. Hovering reveals the exact number of movies for a given year.

Goal: This is an intuitive first point of entry into the dataset. It provides a clear overview of how movie production evolves over time, while allowing users to explore trends for specific genres.

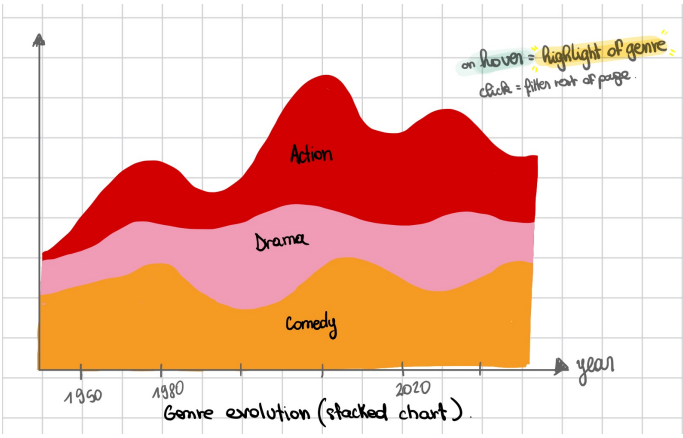
Extra Features: Selected genres can dynamically influence other sections of the website.



2.2 Genre Evolution

Features: This stacked area chart shows how different genres contribute to total movie production over time. Each layer represents a genre, with its thickness indicating its relative importance. Hovering highlights a genre while fading the others, making it easier to track its evolution.

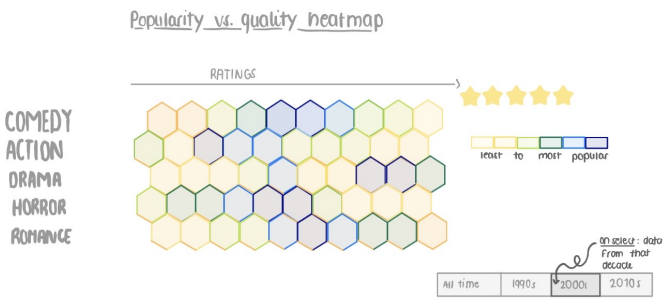
Goal: To show how the distribution of genres changes over time and identify which genres dominate different periods.



2.3 Popularity vs Quality Heatmap

Features: This heatmap represents the relationship between movie ratings (x-axis) and genres (y-axis), with color indicating the density of popularity of movies for each combination. Users can select a decade to update the visualization and compare different time periods.

Goal: To explore the relationship between popularity and quality, and how it evolves across decades.



2.4 Budget vs Revenue Bubble Plot

Features: This bubble plot shows the relationship between budget (x-axis) and revenue (y-axis), with each bubble representing a genre and its size indicating its revenue. A time slider allows users to select a time window, updating the chart dynamically. Hovering reveals detailed values, and clicking shows the top 20 movies by return on investment for the selected genre and period.

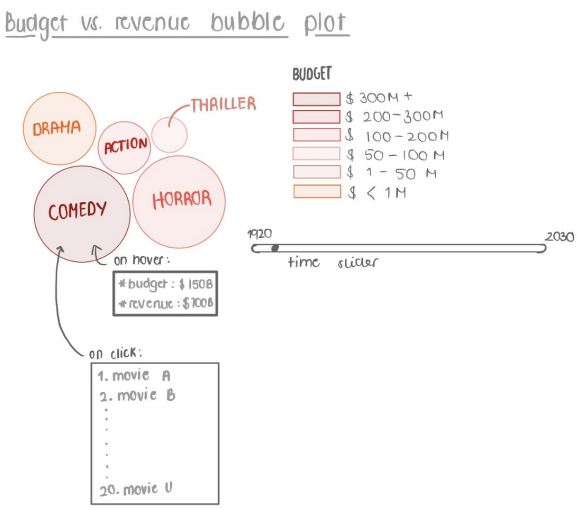
Extra Features: A visual transition from black-and-white to color reflects the historical evolution of cinema. Outliers can be highlighted.

Goal: To analyze how financial investment relates to revenue and compare profitability across genres and time.

2.5 Quality of Movies Through Time

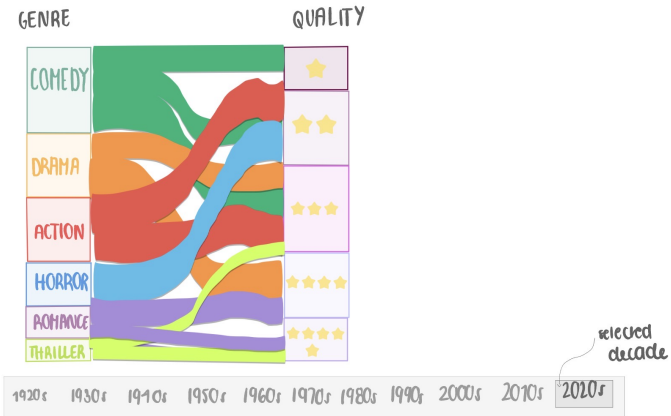
Features: This visualization uses a Sankey chart to show how movies from different genres are distributed across rating categories. The width of each flow indicates how many movies from a genre fall into a given rating group. Time is shown by clicking on a decade and selecting it.

Extra Features: To match the historical narrative of cinema, the visualization begins in black-and-white for decades before 1940 and gradually transitions into color in later periods. We can also make the



time dimension be controlled through a playable animation, allowing the chart to evolve like a short a video rather than manually clicking.

Goal: To show how movie quality is distributed across genres and how this relationship changes over time.



2.6 Representative Movies Panel

Features: This section displays a selection of representative movies based on the currently active filters (time range, genre, etc.). Movies are shown as cards or posters for visual appeal (containing key metadata). Clicking a movie opens its IMDB page.

The goal of this visualization is to connect abstract trends with concrete examples. By showing actual movies, users can better interpret the data and relate insights to familiar content. This section also enhances engagement by making the experience more tangible and visually rich.

3 Tools & Lectures

The website will be implemented using modern web technologies. The lectures and tools used for each visualization are summarized below:

Purpose	**Tool**
Layout & styling	HTML / CSS (Flexbox & Grid)
Interactivity	JavaScript
Version control & CI/CD	GitHub Pages (auto-deploy from main)
Data format	JSON

Table 2: Cross-cutting tools used in the project

Visualization	Tools/Libraries used	Course lectures referenced
Production over time	D3.js (line chart) JavaScript ES6 (filtering, hover interactions)	Tabular Data: Simple line chart, Showing trends Interaction: Filtering, Details on demand
Genre evolution	D3.js (stacked area chart) JavaScript (hover highlight)	Tabular Data: Stacked area chart Interaction: Focus + context
Ratings & popularity over time	D3.js (line charts) JavaScript (toggle, filtering)	Tabular Data: Simple line chart, Showing trends Interaction: Filtering, Details on demand
Budget vs Revenue	D3.js (bubble plot) JavaScript (filtering, hover interactions)	Tabular Data: Scatterplots, Bubble plots Interaction: Filtering, Details on demand
Quality of movies through time	D3.js (sankey flow chart) D3-sankey	Maps: Sankey flow chart
Representative movies panel	HTML / CSS (cards layout) JavaScript (dynamic rendering)	Storytelling

Table 1: Visualization tools and corresponding course lectures

What each tool does and why we use it

Tool	What it is	Why we need it in the project
D3.js	JavaScript library for creating custom, data-driven visualizations using SVG and DOM manipulation.	Used to build all interactive visualizations with full control over visual encoding and user interactions.
Vanilla JavaScript (ES6)	Modern JS features (let/const, arrow functions, template literals, modules).	All page logic (JSON loading, UI events, chart updates) is written in straightforward, framework-free code.
HTML / CSS (Flexbox, Grid)	Core web standards for structure and responsive layout.	Ensures the site looks good on laptop or mobile without extra CSS frameworks.
GitHub Pages	Static hosting directly from the repository with automatic rebuild on push.	Free continuous deployment and a shareable prototype URL for reviewers.
JSON (pre-processed)	Lightweight data-interchange format parsed natively by browsers.	All commuter counts are shipped as static JSON; no server or Python needed.

Table 3: Description of the tools